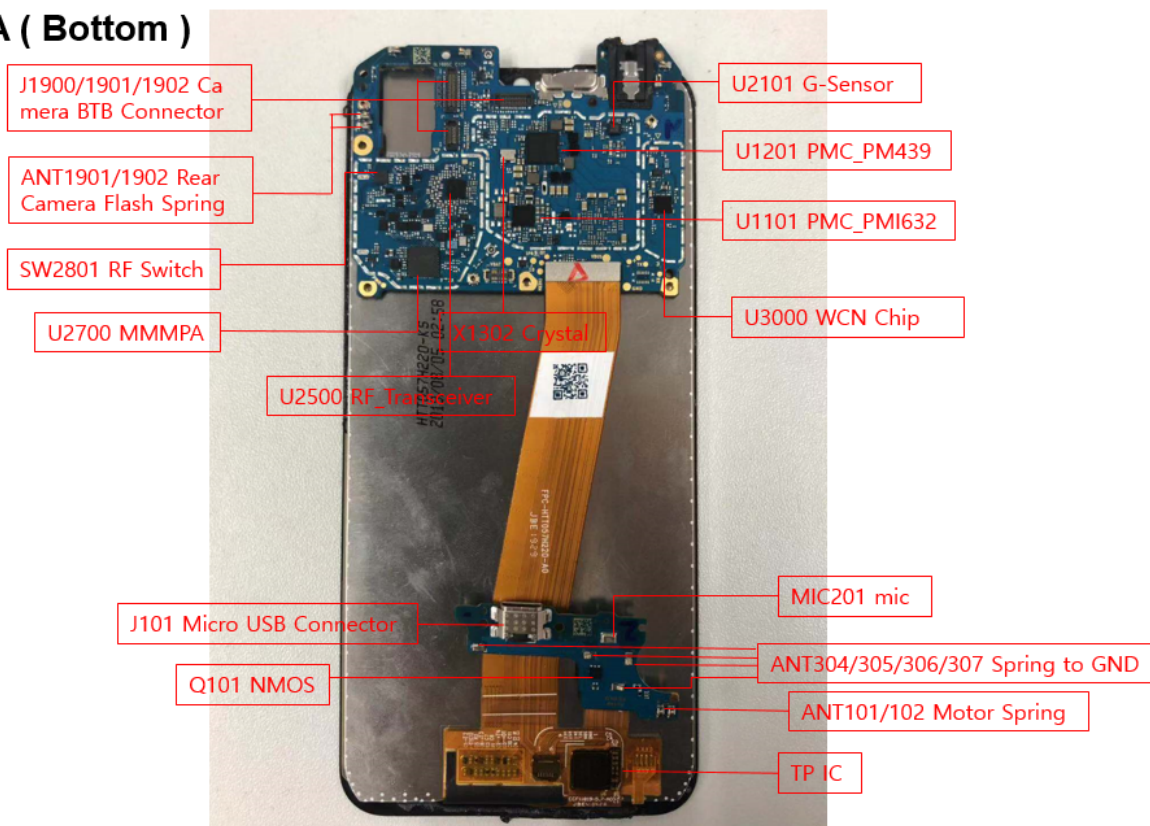
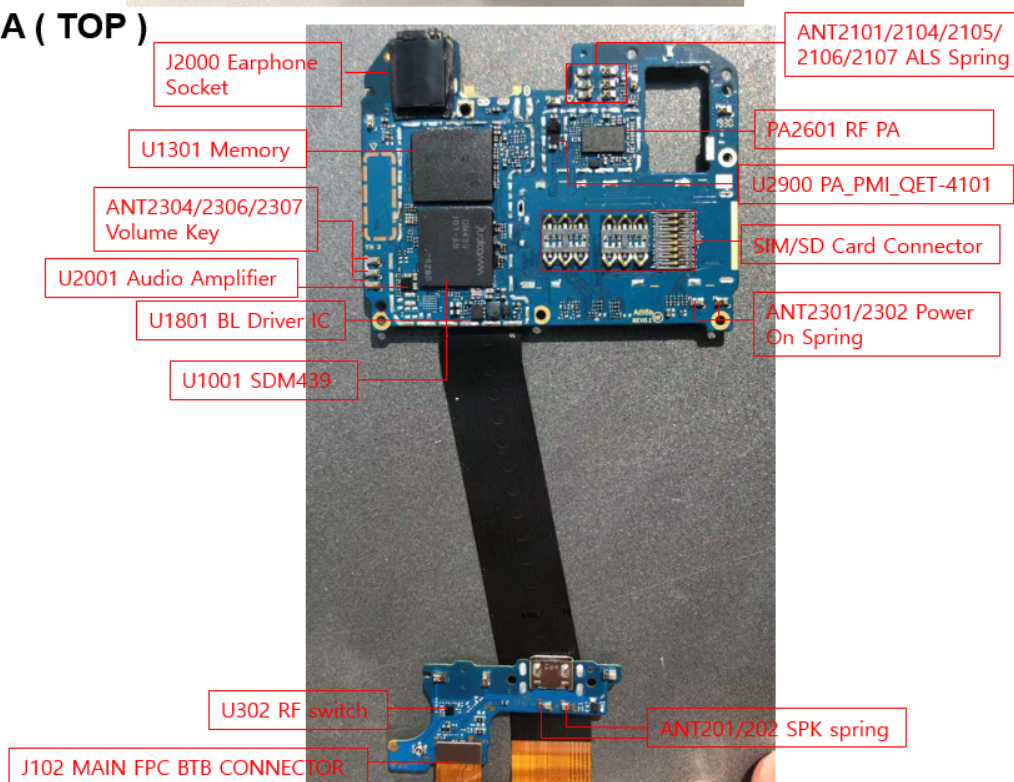

8. Level 3 Repair

8-1. Components Layout

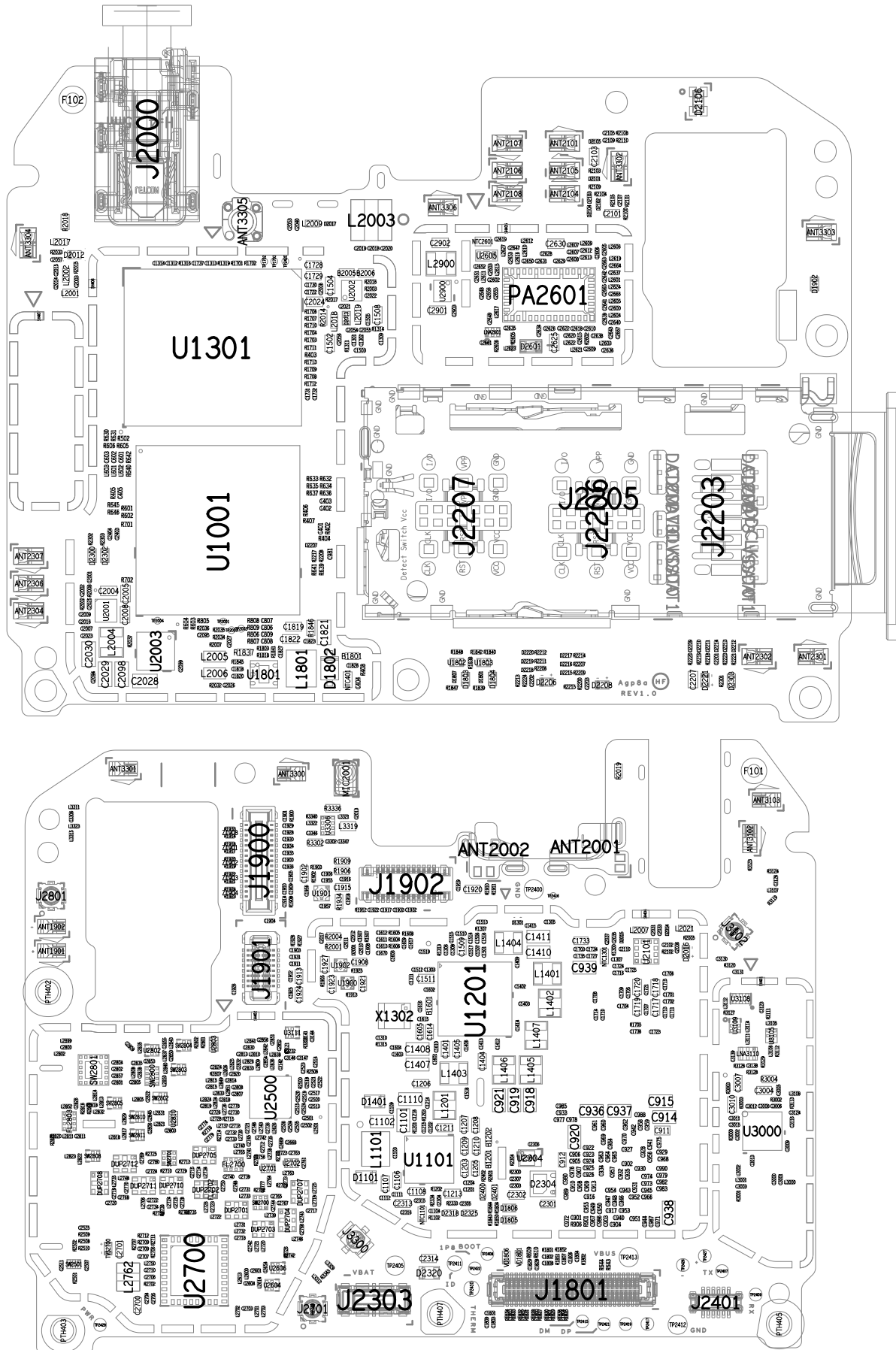
PBA (Bottom)



PBA (TOP)



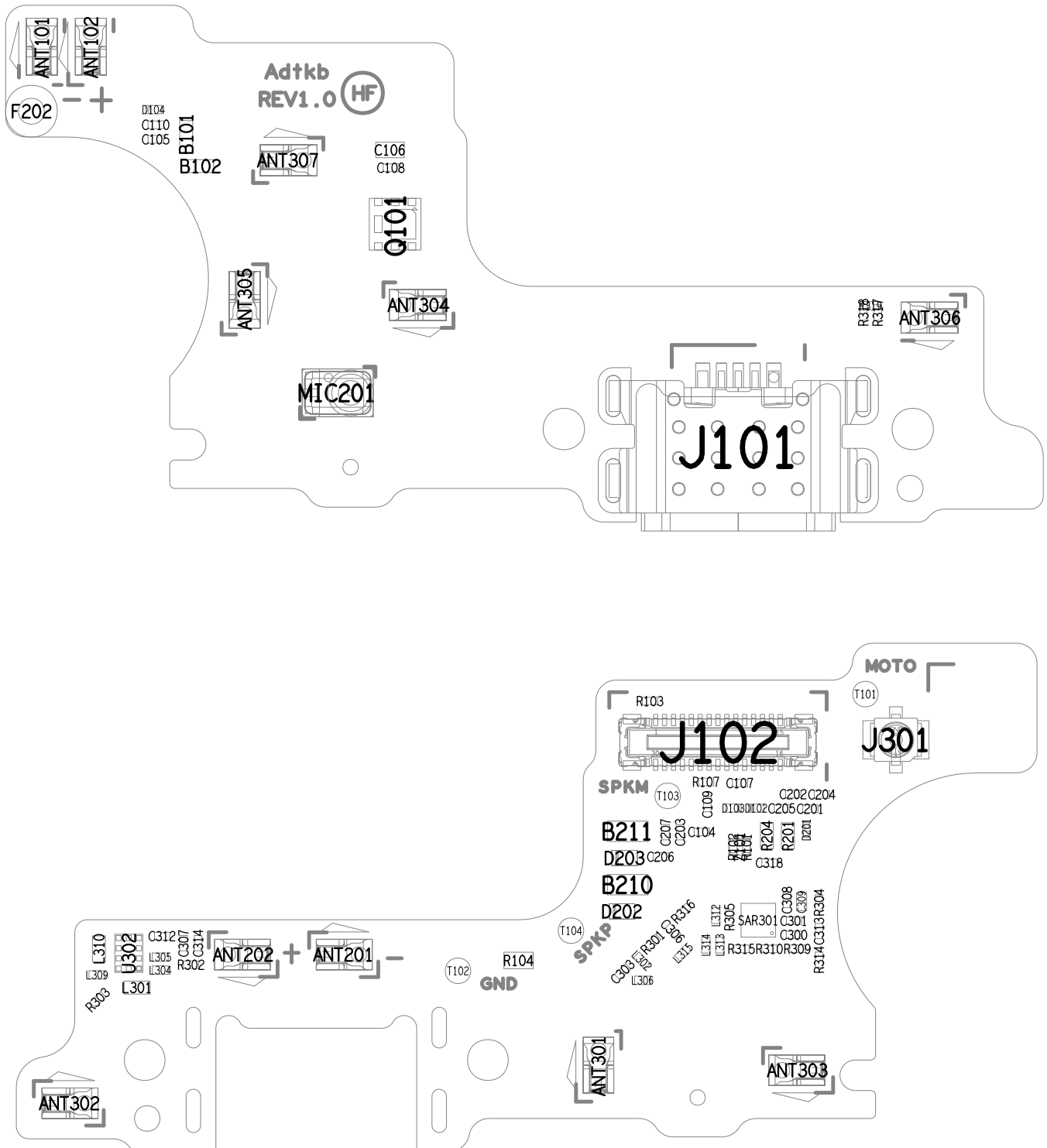
8. Level 3 Repair



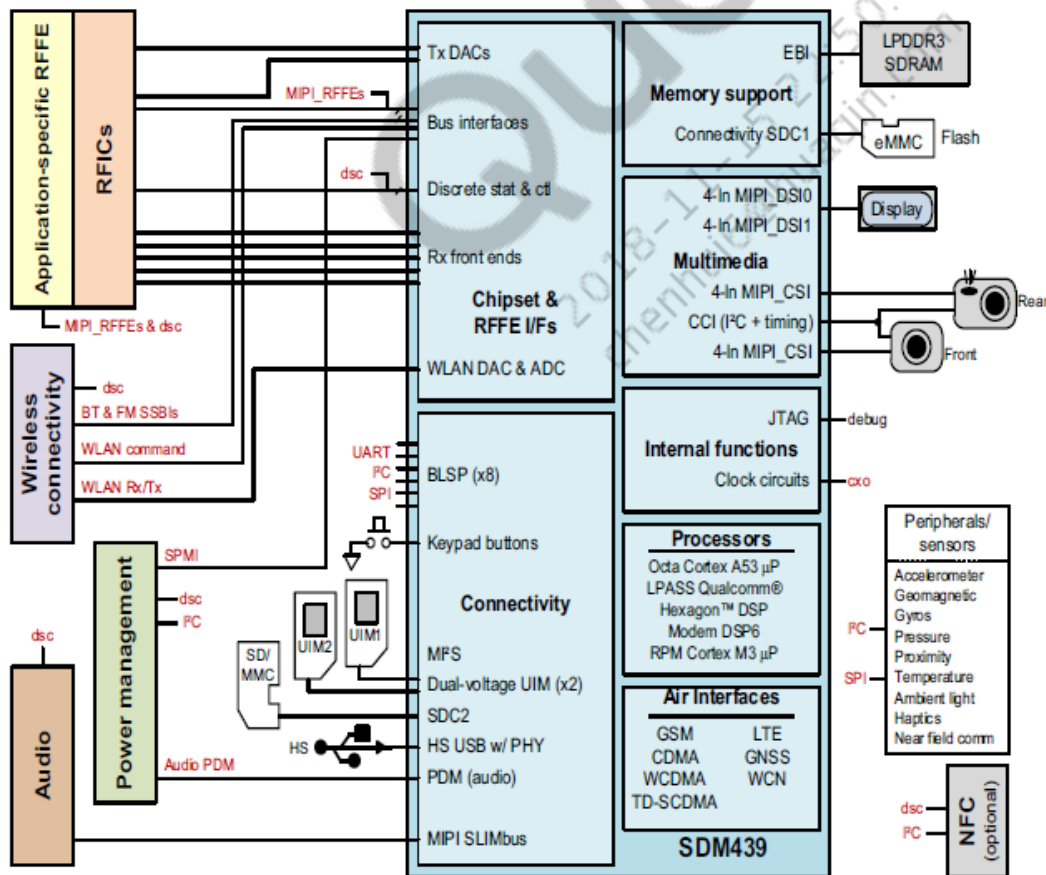
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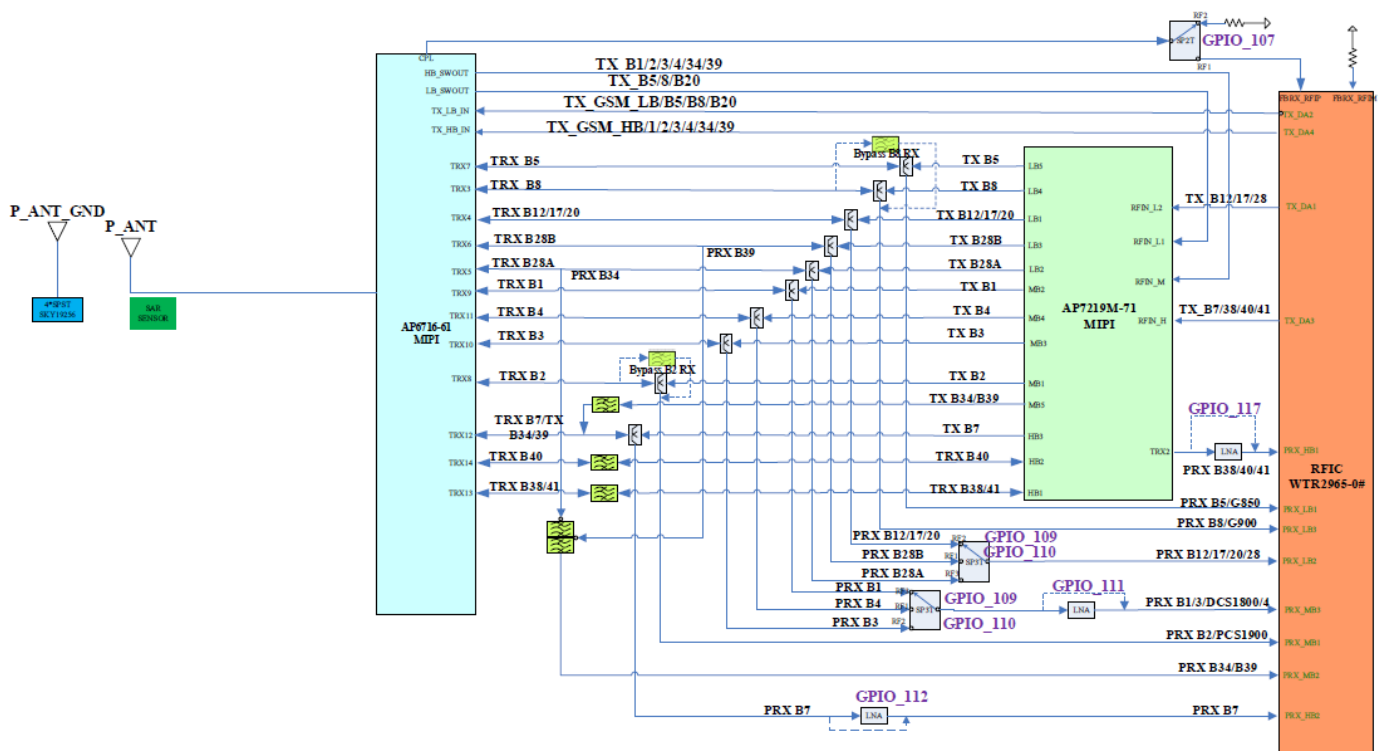
8. Level 3 Repair



8. Level 3 Repair



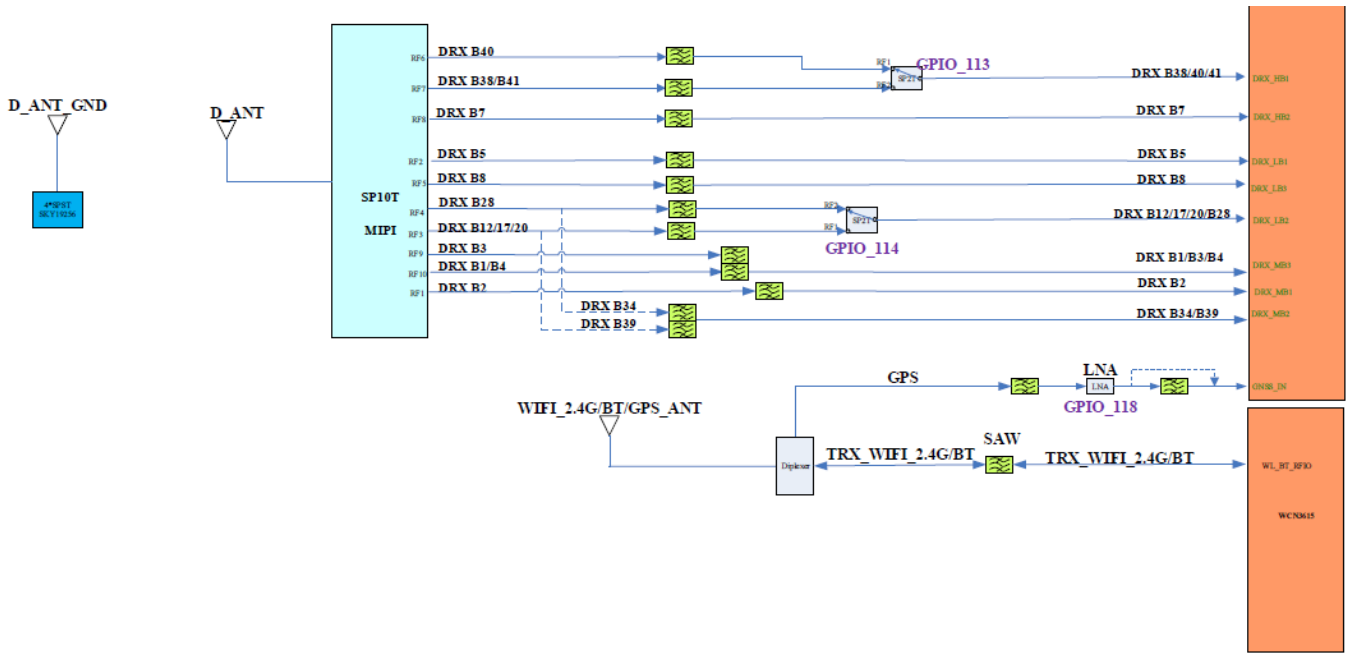
ZQL1695 RF Block Diagram



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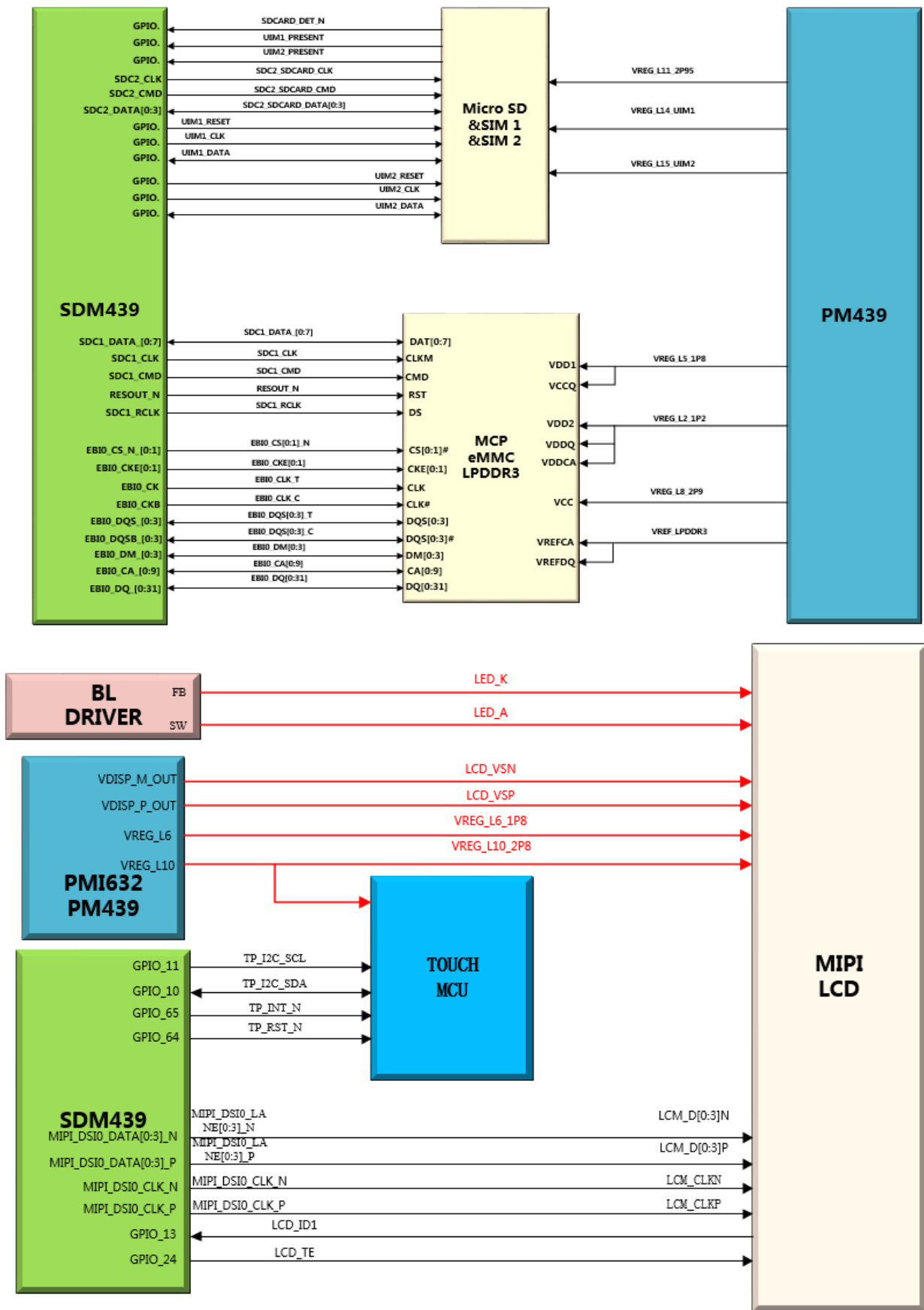
8. Level 3 Repair



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8. Level 3 Repair

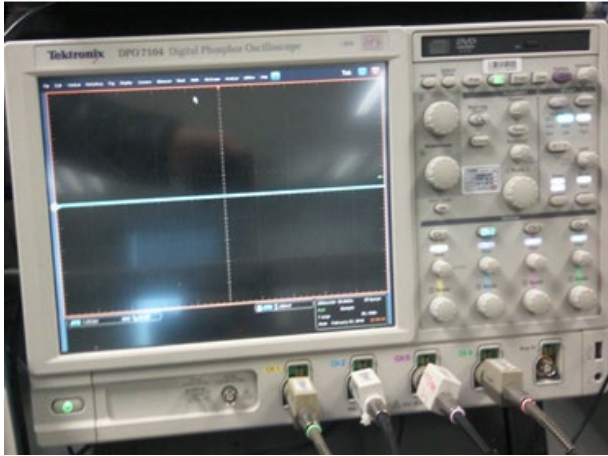


8-3. Flow chart of Troubleshooting.

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8. Level 3 Repair



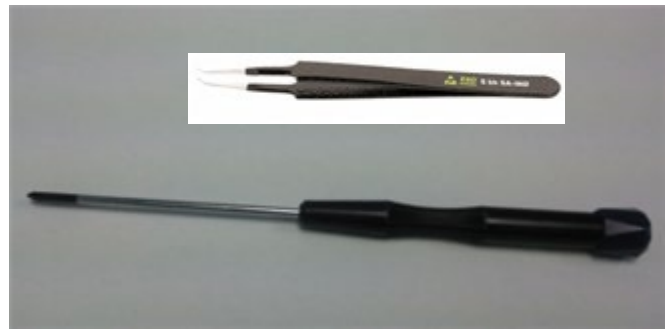
Oscilloscope



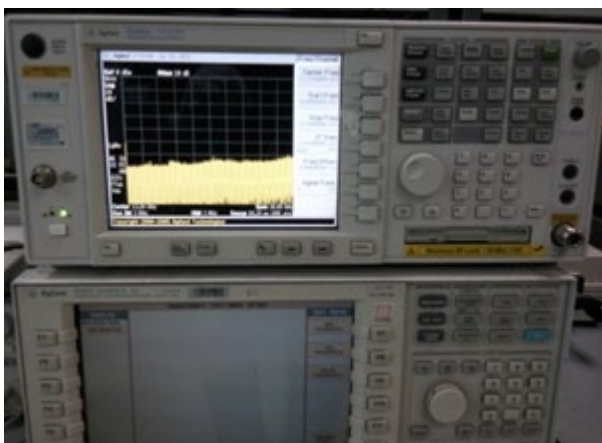
Digital Multimeter



Power Supply



+ driver, ESD Safe Tweezer



8960 & Spectrum Analyzer

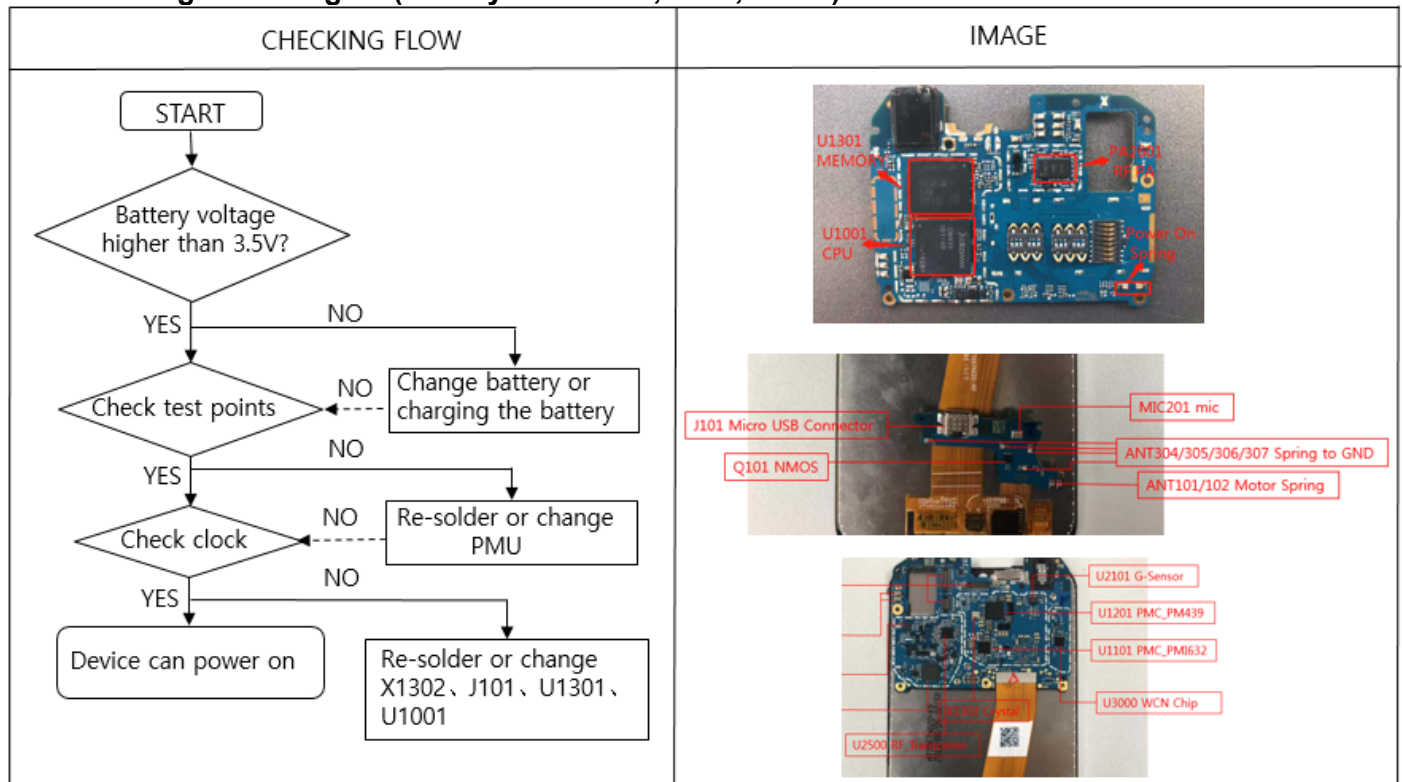


Soldering iron

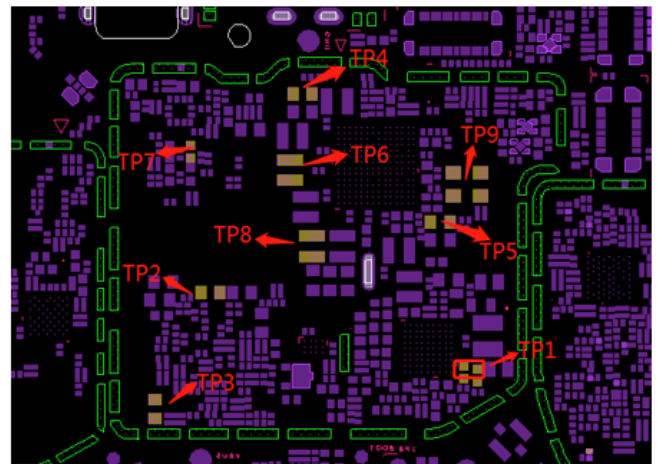
8-4-1. Power On

8. Level 3 Repair

■ Checking Power signal (Battery connector, PMU, Clock)



Power on voltage check				
Power Domain	Configurable Voltage	Signal Name	Measurment Location	TP
VPH_PWR	3.0-5.25V	VPH_PWR	C1106/C1107	TP1
VDD_MEM	0.915V	VREG_S7_0P915	C937	TP2
VDD_CORE	0.8625V	VREG_S1_0P8625	C938	TP3
PM439_LDO	2.05V	VREG_S4_2P05	C1411	TP4
PM439_LDO	1.225V	VREG_S3_1P225	C1408	TP5
PM439_LDO	0.8625V	VREG_S2_0P8625	L1402	TP6
LP-DDR3	1.2V	VREG_L2_1P2	C1720	TP7
APC	0.862V	VREG_S5_S6_0P8625	L1405	TP8



➤ Oscillate frequency measurement

Signal Name	Frequency MHz	Measurment Location	TP
XTAL_19M_IN	19.2M	X1302	TP9

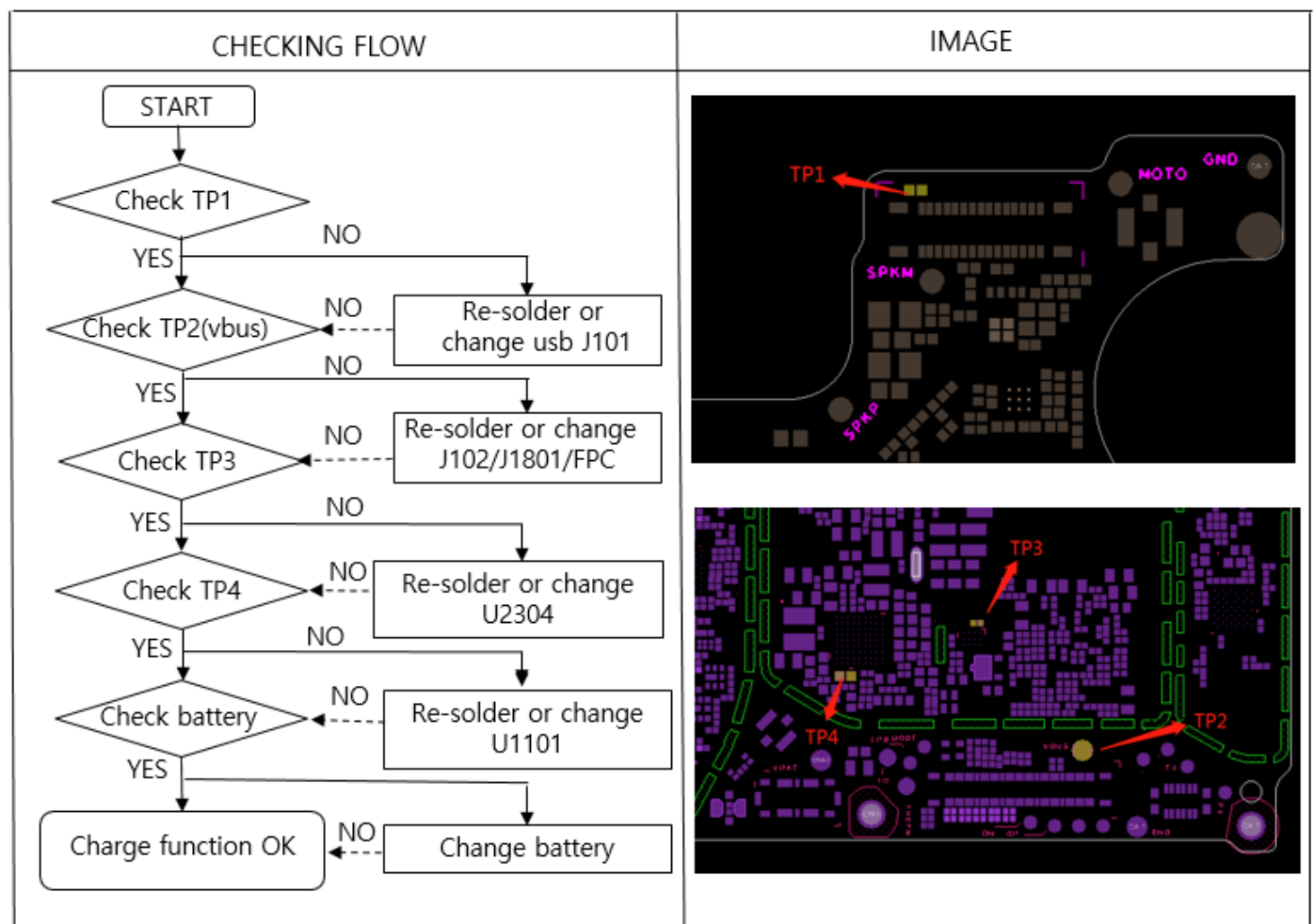
8-4-2. Charging

■ The charging controlled by PMU chip PM660 (U1501)

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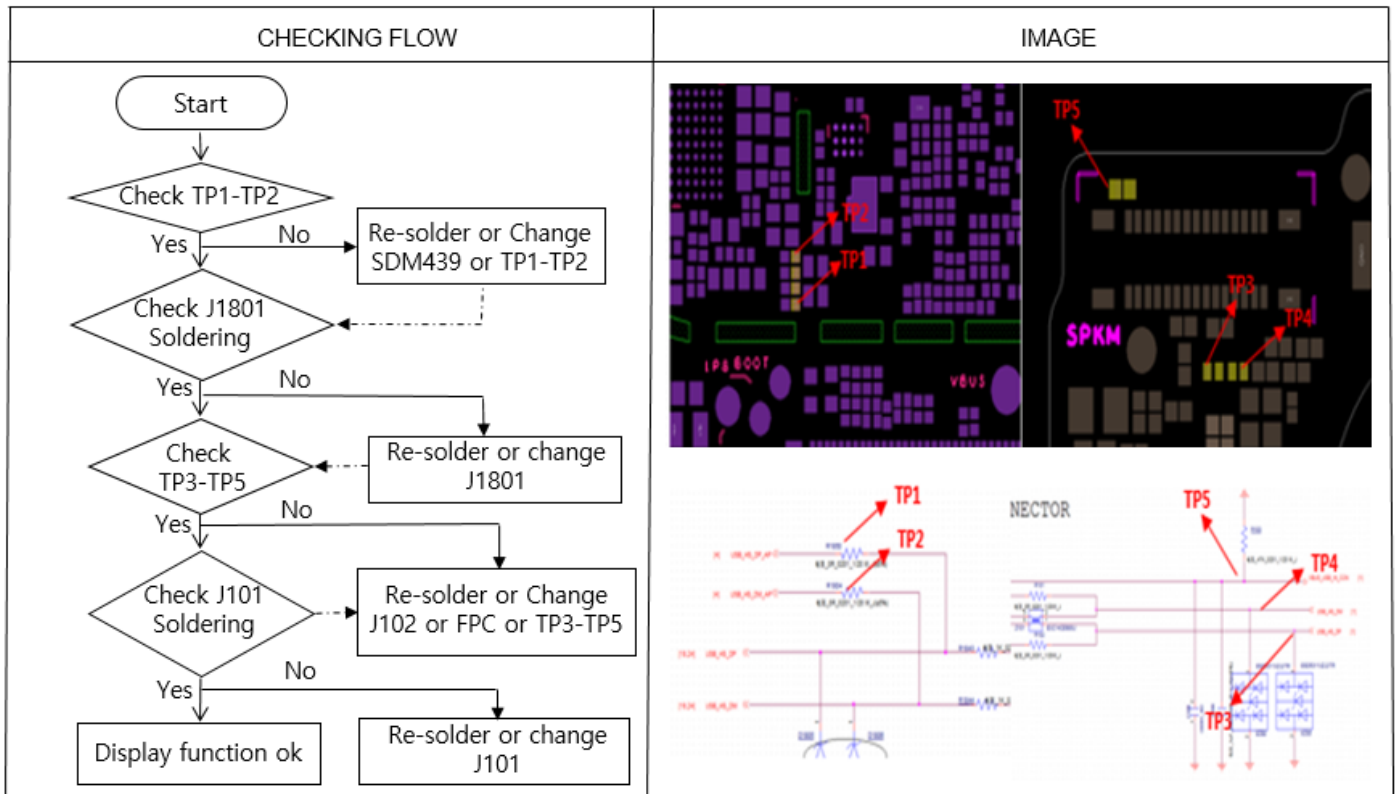
8. Level 3 Repair



8-4-3. USB

- I/O connector is used as the USB port.

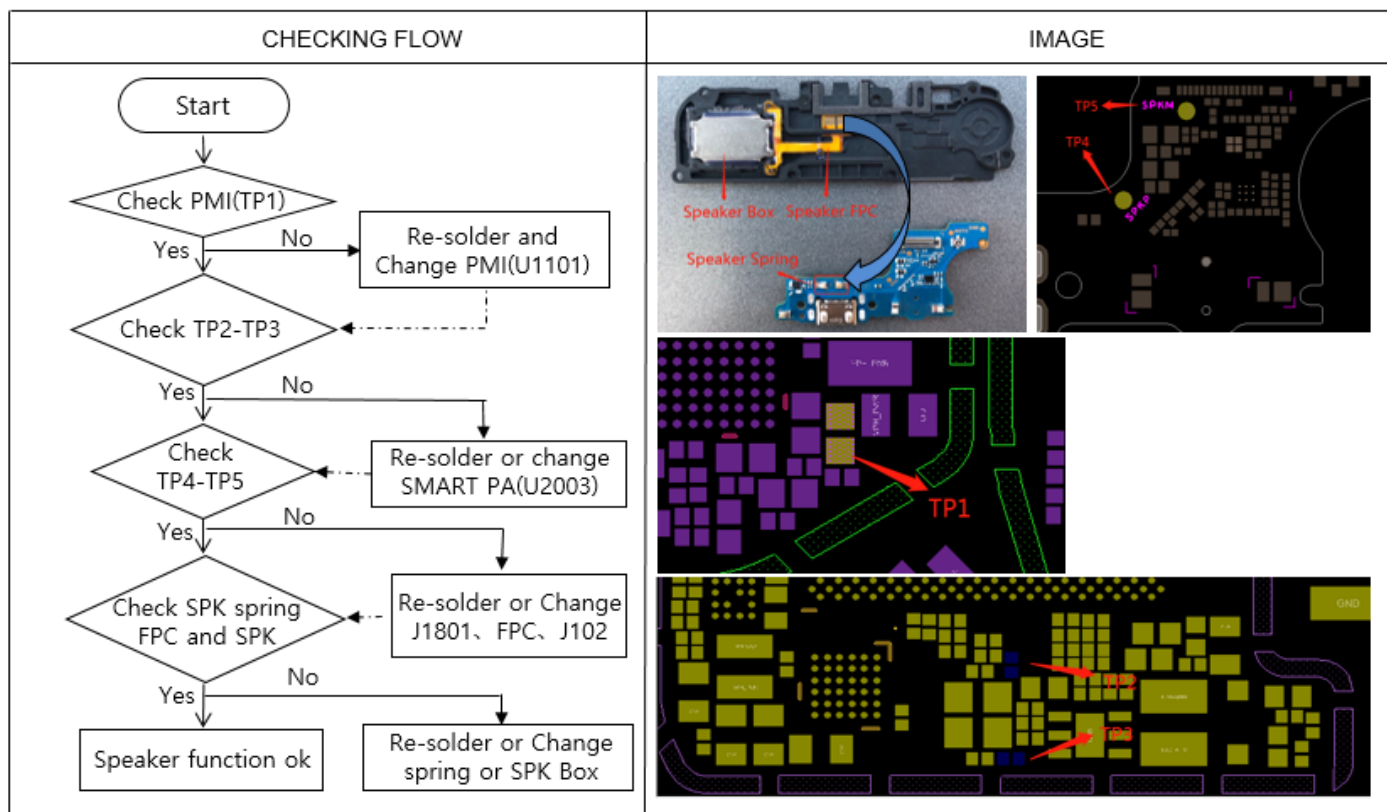
8. Level 3 Repair



8-4-4. Audio speaker

- The Speaker control signals are generated by BB chip SDM439(U1001) and SMART PA, the chip, PA and the speaker are to be checked out.

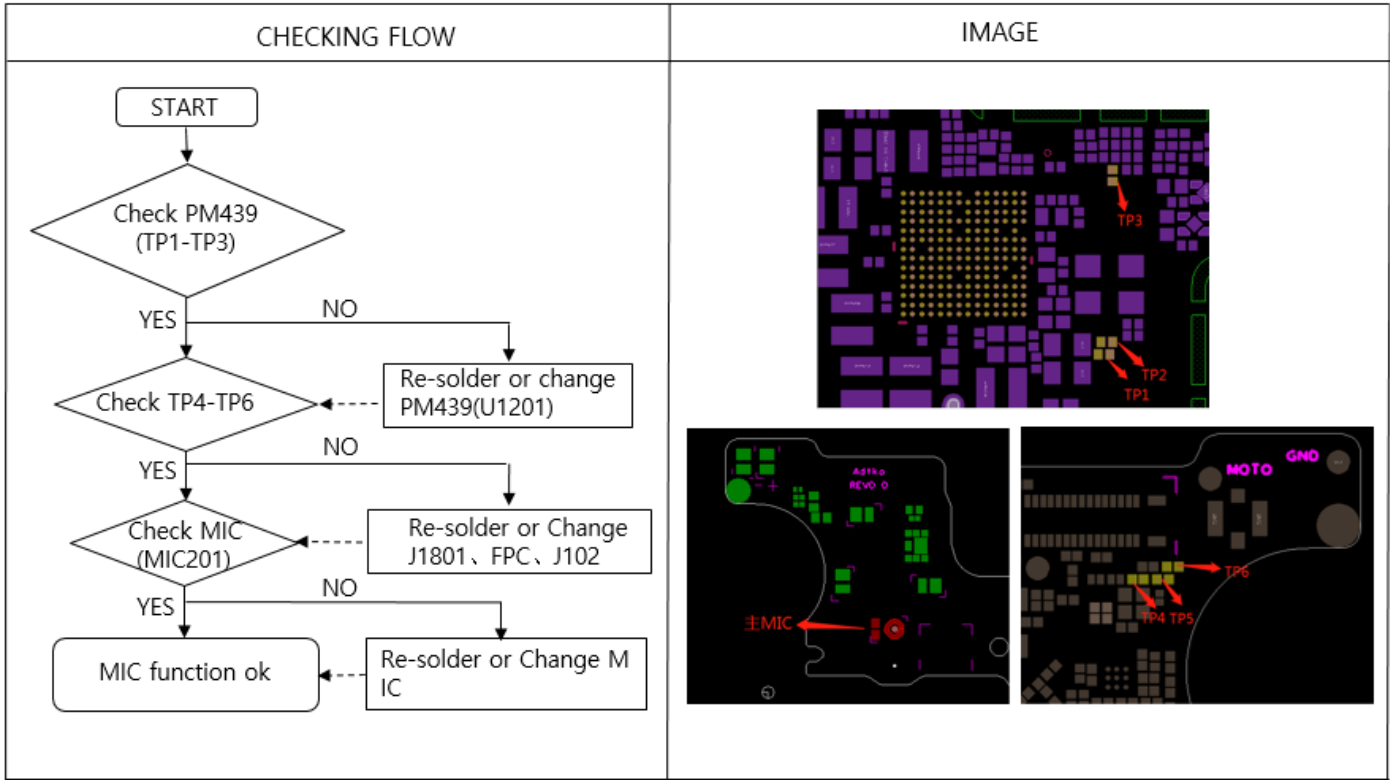
8. Level 3 Repair



8-4-5. Audio_Main MIC

- The MIC control signals are generated by PMU chip PM439(U1201), the PMU chip and the MIC are to be checked out.

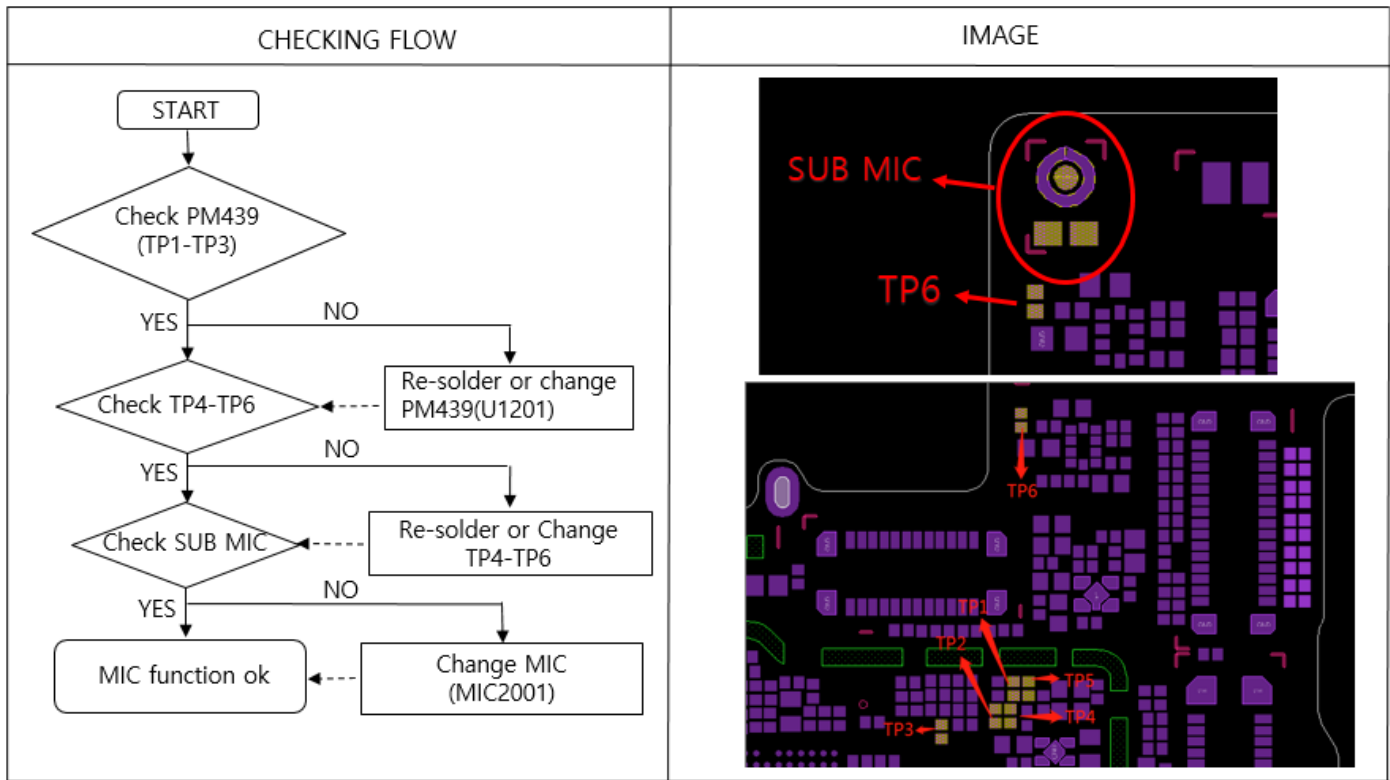
8. Level 3 Repair



8-4-6. Audio_Headset MIC

- The MIC control signals are generated by PM439 (U1201), the PMU chip and the MIC are to be checked out.

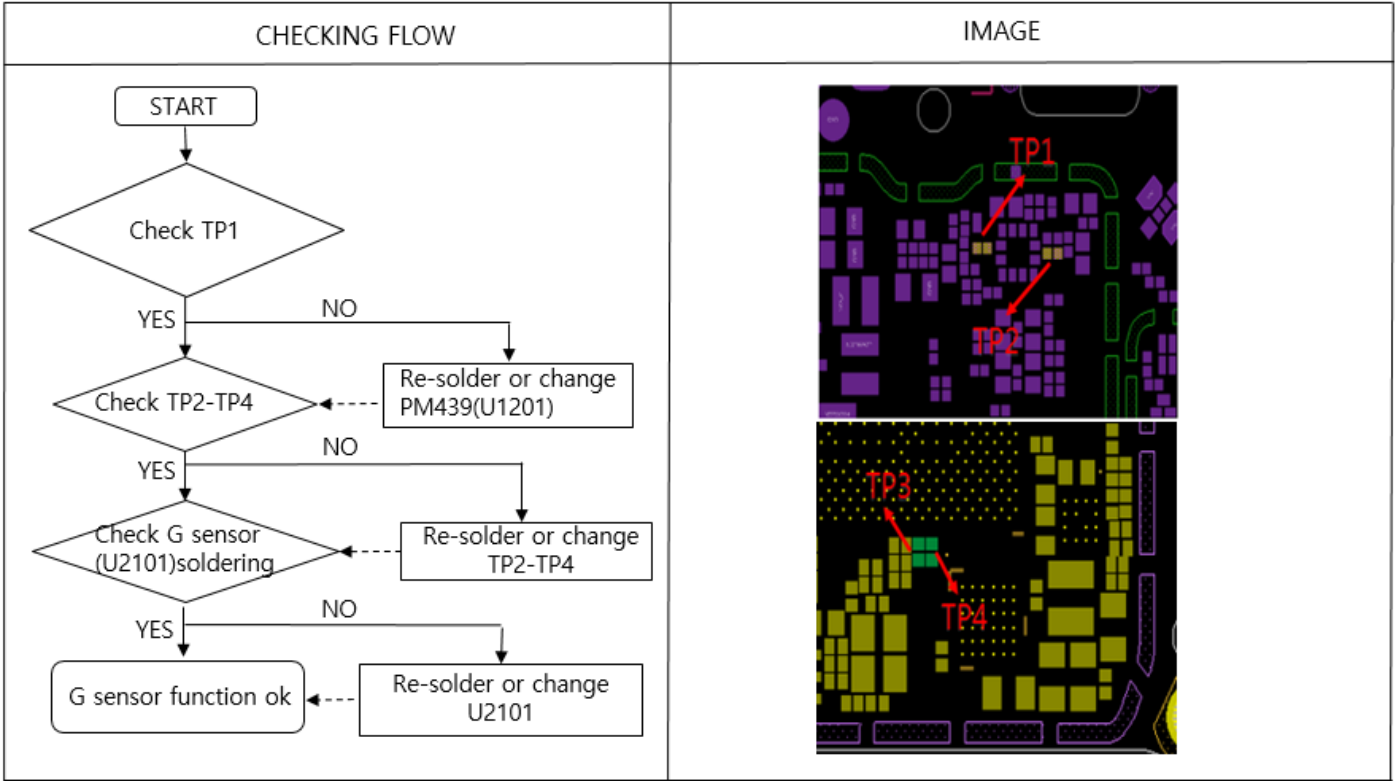
8. Level 3 Repair



8-4-7. G sensor

- The G sensor is calibrated by using SW algorithm.

8. Level 3 Repair

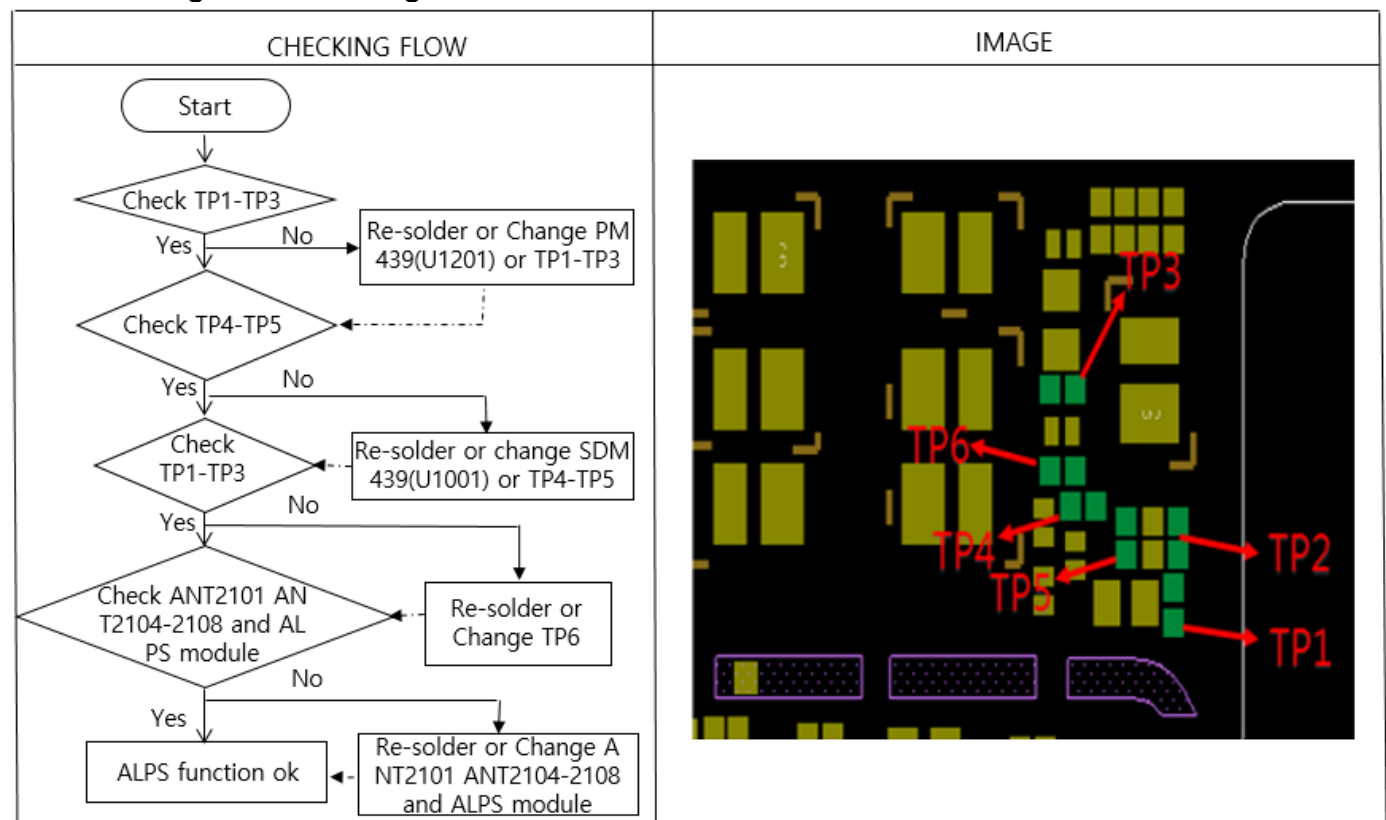


8-4-8. Proximity and light sensor

- Proximity and Light Sensor is worked as below: Control the screen's on/off operation automatically while making phone calls, and adjust the screen brightness

8. Level 3 Repair

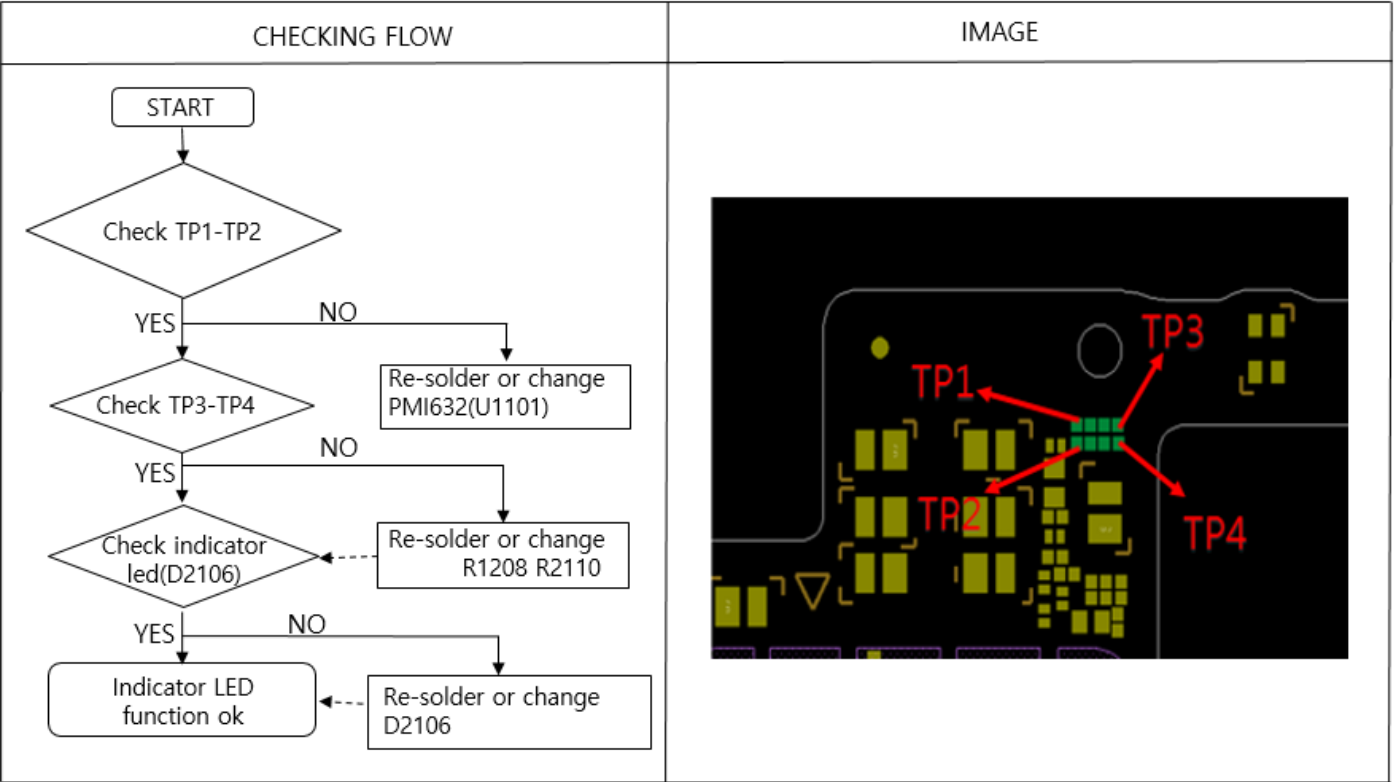
according to ambient light.



8-4-10. INDICATOR LED

- INDICATOR LED is worked as below: Information cue for charging, phone call and message

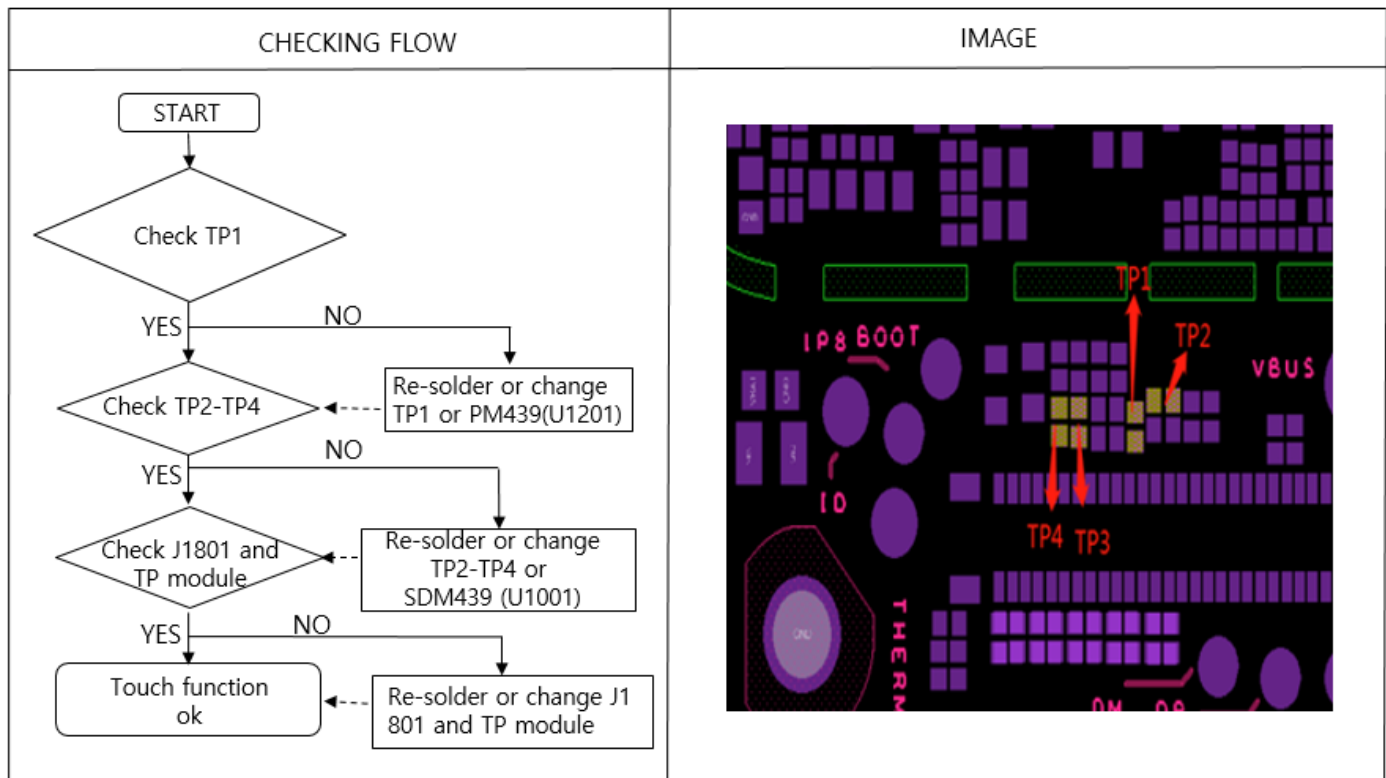
8. Level 3 Repair



8-4-11. TOUCH SCREEN

- The Touch control signals are generated by SDM439 and SDM439. It is assembled with LCD.

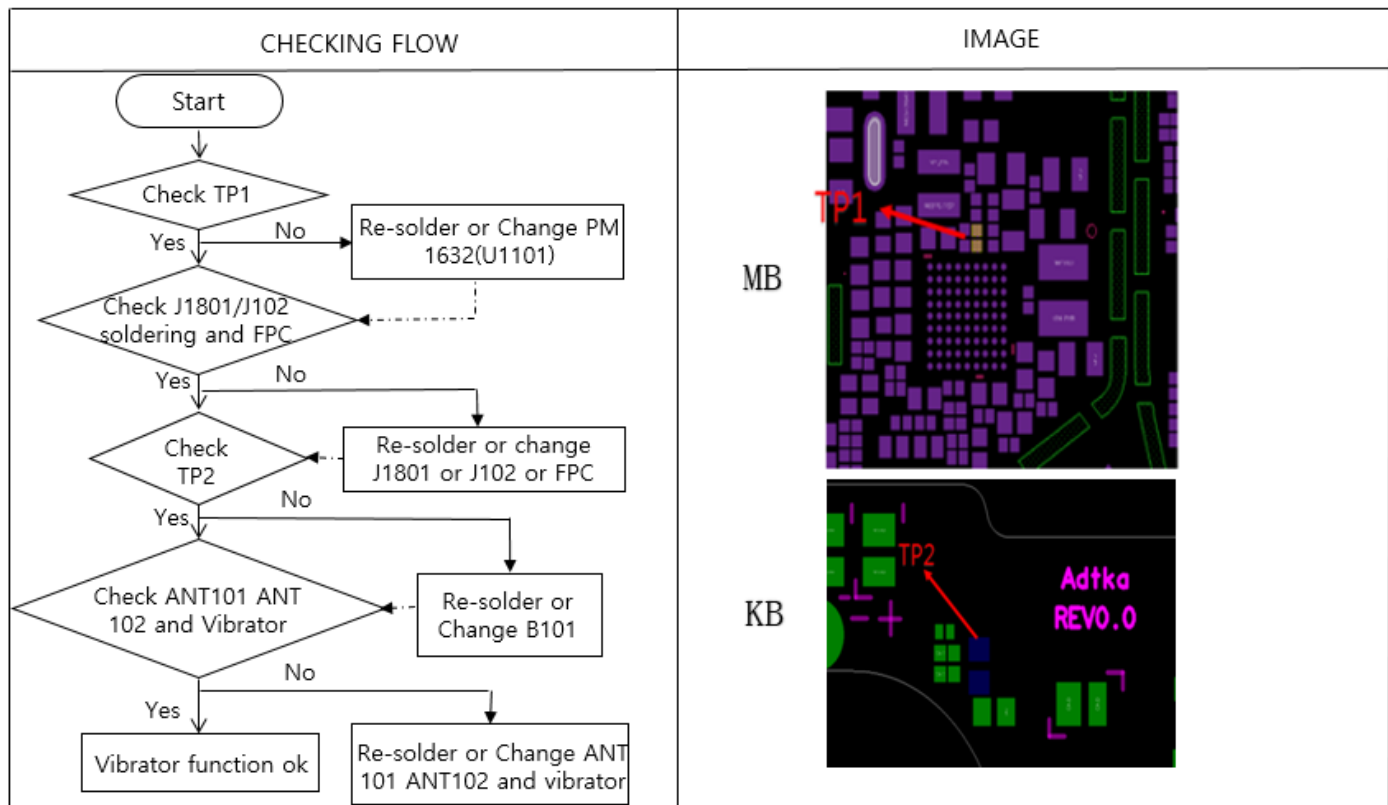
8. Level 3 Repair



8-4-12. Vibrator

- The Vibrator control signals are generated by PMI632.

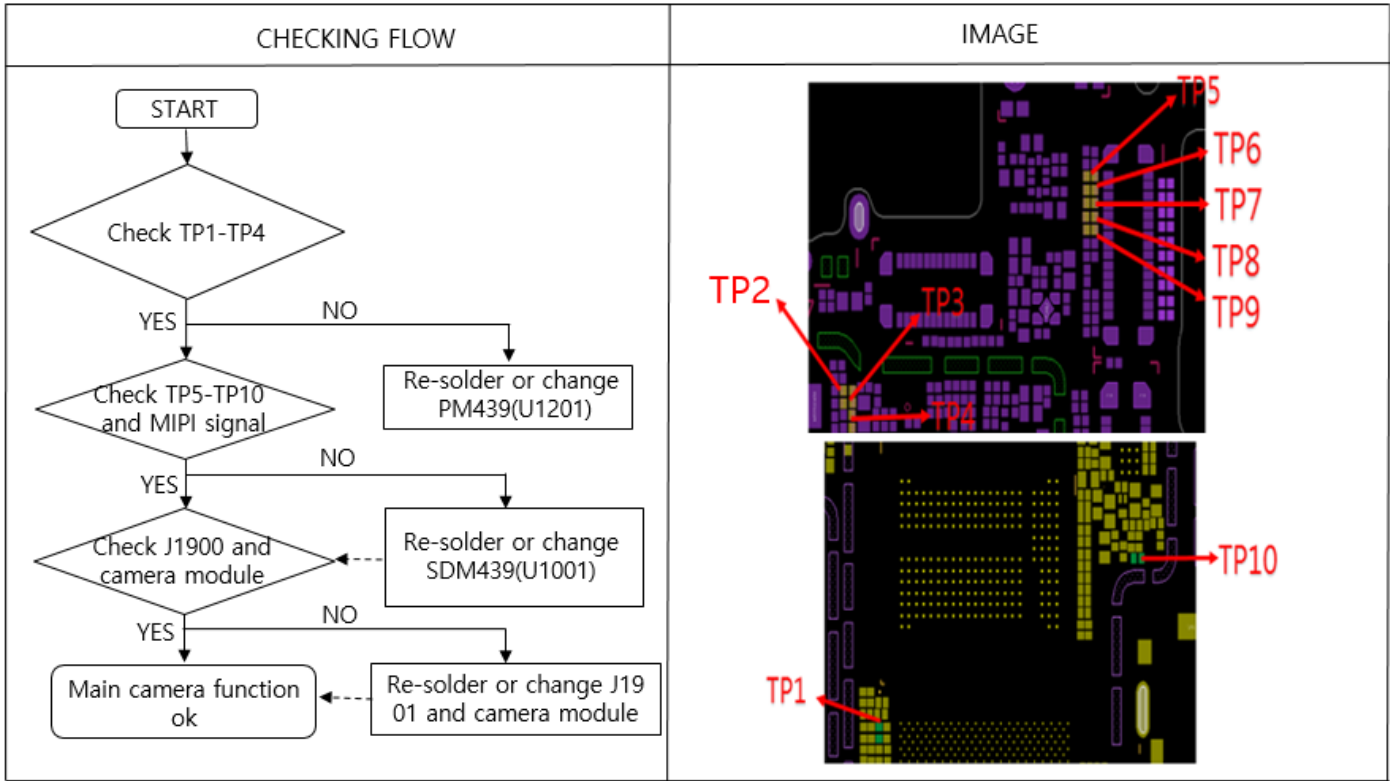
8. Level 3 Repair



8-4-13. Main Camera

- The Camera control signals are generated by PM439 and SDM439.

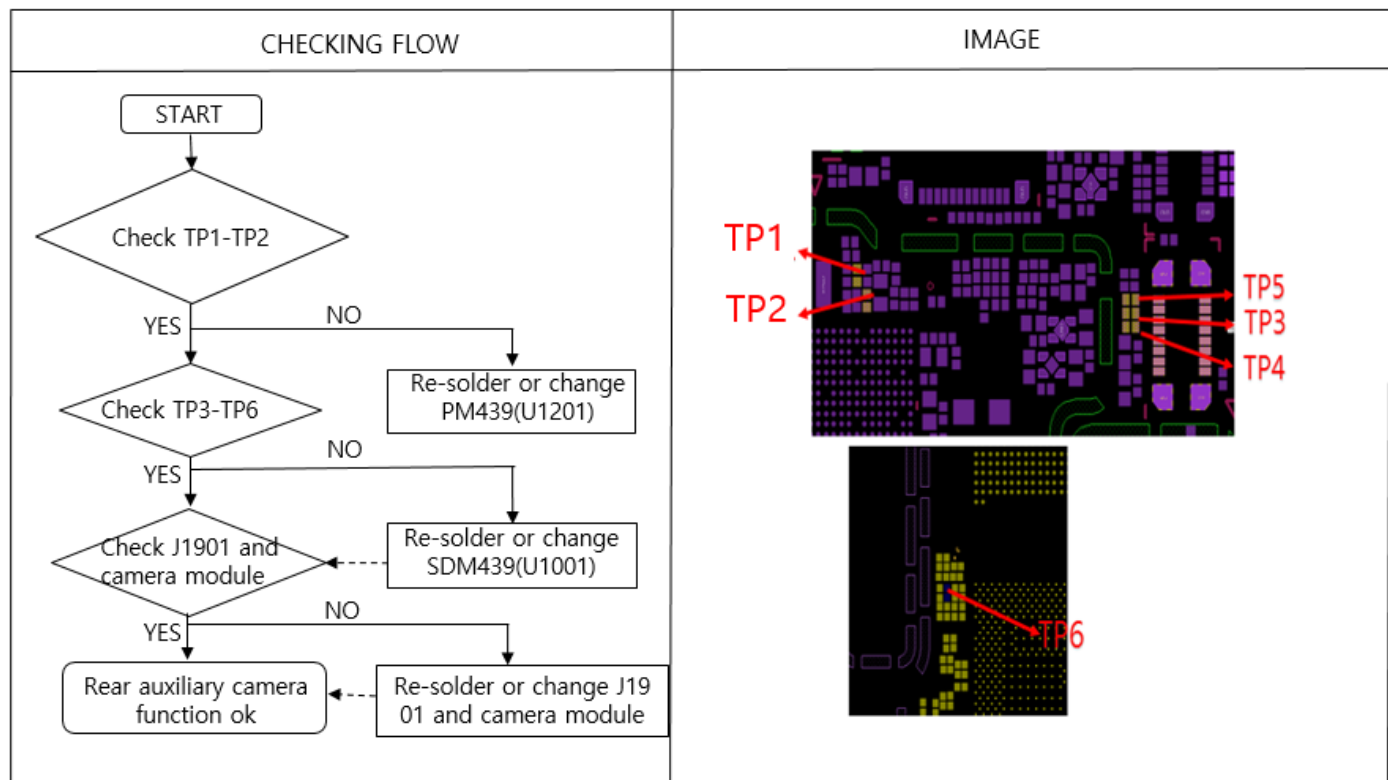
8. Level 3 Repair



8-4-14. Rear auxiliary Camera

- The Camera control signals are generated by SDM439 and PM439.

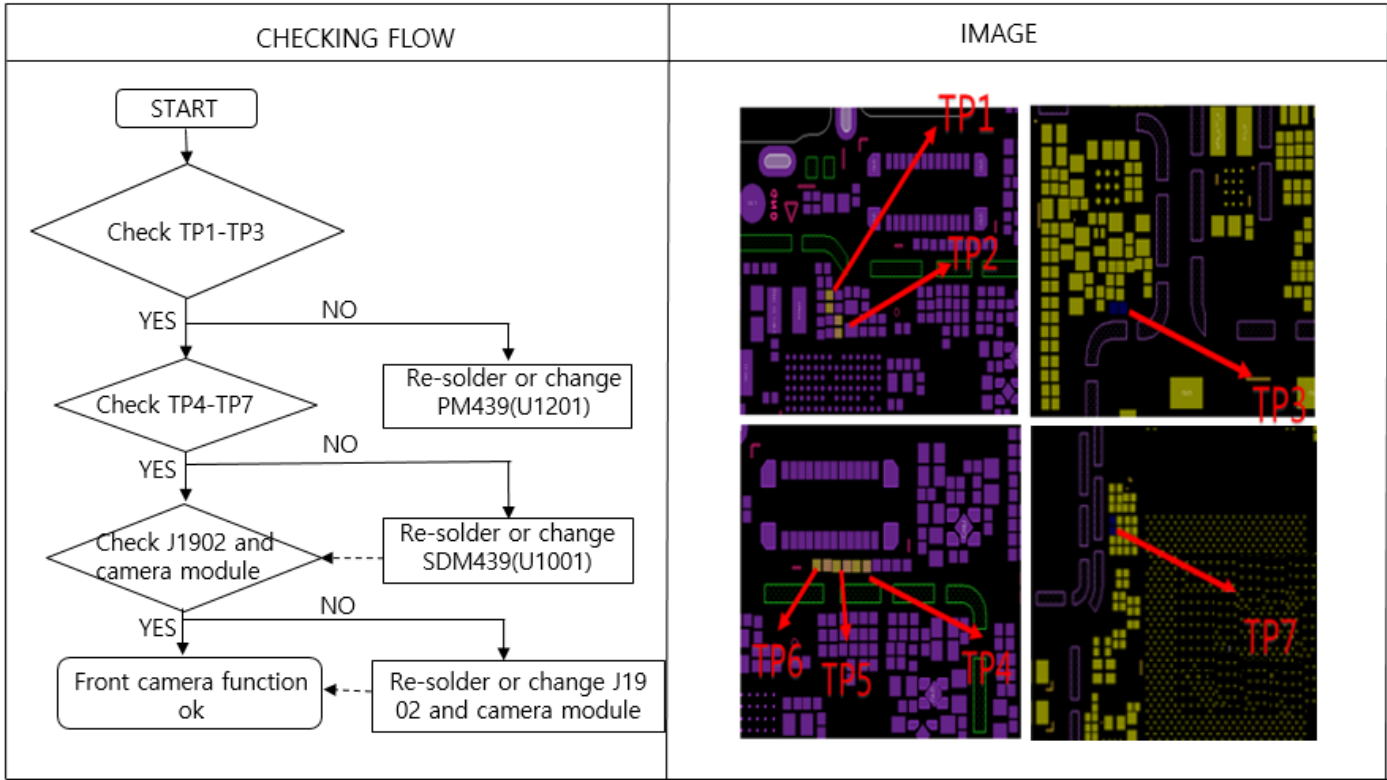
8. Level 3 Repair



8-4-15. Front Camera

- The Front Camera control signals are generated by PM439.

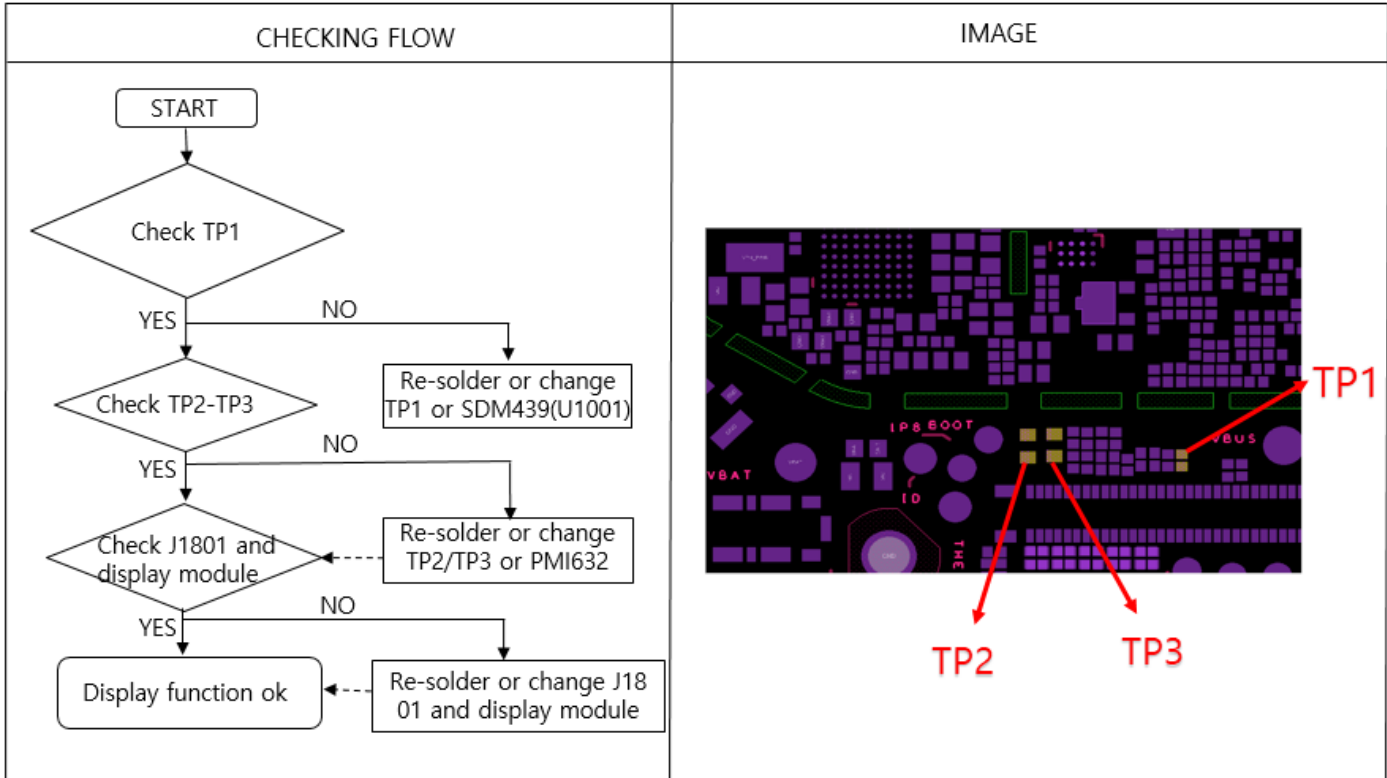
8. Level 3 Repair



8-4-16. LCD

- The LCD control signals are generated by SDM439.

8. Level 3 Repair



8-5. Service Schematics

■ U1001_SDM439_BB chip IC , Digital Baseband Processor(Top)

